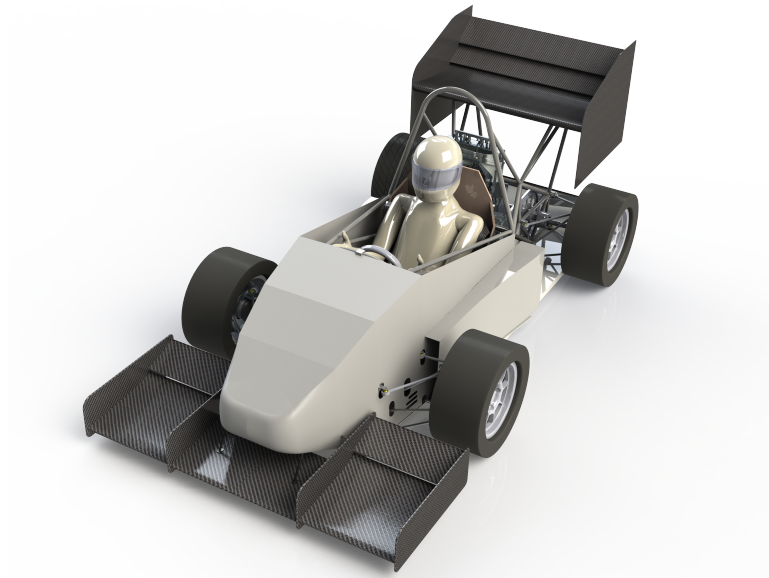


GROUP DESIGN PROJECT
IMPERIAL RACING GREEN

Front Wing Design



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Course: MEng Aeronautical Engineering
Module: AERO60004 Group Design Project
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Abstract

The focus of this report is to present the development of a front wing design to be integrated into the HEV4 prototype, a combustion-electric hybrid car for the Imperial Racing Green team to compete in the 2022 Formula Student competition. The front wing is a key aerodynamic device aiming to improve the grip and handling capabilities of the car through generating up to 40% of the total downforce [1]. The design objective is to create a configuration capable of meeting the 141 N downforce target, with a lift-to-drag efficiency of 3.81, while also improving the flowfield around other components of the car, such as the floor, diffuser, nose cone and cooling system. CFD simulations are used as the main tool within the optimisation process, with the potential increase in competition points quantified for each design decision adopted. A double-element wing configuration is selected for minimising the drag induced, with rectangular endplates and Gurney flaps included for an improvement in performance. The front wing selected configuration is capable of ensuring an L/D ratio as high as 5.17, but for only 119 N of downforce generated, 15.6% lower than targeted. The choice of CFRP to manufacture the device reduces the mass to 4.55 kg, for a cost of 1470£. Overall, the addition of an aerodynamic package provides a 90.9 points gain [2], making it a viable and efficient solution for the car development.



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Executive Summary

Technical Coordination

HEV4 is a hybrid electric Formula Student car with a double wishbone independent pullrod suspension system with a steel spaceframe chassis, a wheelbase of 1624 mm, and an aerodynamic package with a front wing, rear wing and undertray and diffuser. The objective of this group project was to design HEV4 to be a winning car in the Formula Student 2022 competition held at Silverstone for the customer, Imperial Racing Green. This was done by aiming for the maximum number of points from a series of static simulation and business plan and design evaluations, and dynamic events including acceleration, autocross, endurance and efficiency. In order to achieve this objective, design decisions were made by being both conceptually evaluated and validated through the use of a lap simulator, all whilst ensuring the car was legal to compete according to the 2022 Formula Student UK regulations.

For more information, please refer to Technical Coordination by Megan Gilkes

Vehicle Performance

Within the scope of this project the role of the Vehicle Dynamics and Performance team was to facilitate other design teams in quantifying their decision making, and to drive the design direction of the car from the very start. The most appropriate method during a design process to identify trends and optimise performance is using a lap time simulator tool, which will then also allow for the conduct of sensitivity analyses and lead the design teams in the importance of their parameters. The Imperial Formula Student Lap Simulator (Imperial Lap Sim) was the main lap sim used during this project. This was verified using a commercial lap time simulator known as OptimumLap. Due to the nature of computational simulation it was first necessary to validate both simulators which also allowed the subteam to identify the strengths and weaknesses of each lap sim.

Problems identified with the Imperial Lap Sim were addressed and the simulator was modified based on the needs of this project such as the hybrid powertrain, as well as some of the needs of other sub teams. The updated Imperial Lap Sim was then revalidated and an improvement in accuracy was observed from the validation results. The lap time simulator was used to obtain results for other subteams which allowed them to make quantitative design decisions. Sensitivity analyses were conducted which allowed sub teams to prioritise each of their parameters. Finally, the vehicle parameters of the final design were used to obtain a performance prediction, which places HEV4 in 1st in the 2021 Formula Student competition with a predicted actual points score of 540.9, and a 62.6% confidence level that the team would obtain 1st.

For more information, please refer to Vehicle Performance Simulation by Xinyu Su, and Vehicle Dynamics and Performance Simulation by Dominic Williams

Global Vehicle Layout

The layout and packaging of the vehicle was optimised for performance by points prediction. Initially, the smallest allowable wheelbase of 1525 mm, front track of 1230 mm and rear track of 1200 mm was used. After several iterations with packaging to accommodate the required components, the final wheelbase was calculated to be 1624 mm. Additionally, to accommodate the maximum steering angle, the front track was increased to 1343 mm. The key components of the car include the hybrid powertrain, chassis, suspension and aerodynamic package. The sizing of these components resulted in an overall car mass of 262 kg. The overall centre of gravity (CG) of the car was calculated by taking the weighted average of the centre of gravity of each component. This resulted in a final CG height of 320 mm above the ground, passing the tilt test which imposed a maximum of 346 mm. The resulting static weight distribution was 56.8:43.2 on the front and rear respectively. This improves front grip, favouring the tight corners present in the dynamic events of the competition.

For more information, please refer to Global Vehicle Layout & Weight Balance by Ming See

Chassis

The chassis of the car is responsible for the overall structural integrity and safety of the driver, whilst satisfying the rules and optimised for minimal weight. As the available budget is limited, the baseline configuration was determined to be a welded steel spaceframe over a carbon fibre composite monocoque design. To maximise buckling, bending and displacement resistance, while achieving low weight and cost, stiffness-cost and stiffness-density ratios were used as material merit indexes. AISI 1015 steel, with excellent welding processability was selected as the main material. The design process consisted of the packaging the different components whilst the regulatory constraints. The chassis was composed around three main load cases: roll hoop loading in case of tip over; 3 point ground contact resulting in axial torsion; and frontal impact up to the maximal 40g declaration. A load factor of 2 was selected for dynamic load studies. To meet these load constraints, the following design features were used: triangulation with nodes placed at internal and external loading point to avoid bending of members; continuous load path arrangement; and cross section design based on tension, bending, and buckling theory. Design is supported with Finite Element studies of the given load cases, justified with analytical methods and mesh convergence studies.

For more information, please refer to Chassis Design by Marton Vigh

Suspension

A racing car generates grip through four contact patches with the road, these are connected to the car with suspension. Suspension design is critical for performance, and must achieve the performance gains without structural failure in competition. Double wishbone pull-rod suspension was designed for the HEV4 at both front and rear. Pull rod suspension was favoured over push rod due to the lower associated centre of mass, in a competition where mechanical grip is crucial. Spring rates and damping constants were determined within the constraints of vehicle sprung and unsprung mass, braking performance and ride height. Damping constants for high and low frequency bump and rebound were independently set. Both front and rear anti-roll bars are incorporated, allowing for softer springs and improved road holding capacity to ensure the tyre contact patch grip is maximised throughout the lap. Roll rates of $1.2^\circ/\text{g}$ at the front and $1.3^\circ/\text{g}$ at the rear have been achieved with wheel rates of 28.4mm and 30.0mm front and rear. The highest loaded outside front tyre is setup to be at optimum camber for lateral grip in cornering. Loads for the suspension arms were calculated using the worst case vertical, longitudinal and lateral contact patch loads, with suitable safety factors. A set of 4-way adjustable *Eibach* springs and *GAZ* dampers will be purchased at a cost of £812. Adjustability of camber, ride height, wheel rate, damping constants and roll stiffnesses allows the driver to tune the performance characteristics of the car to their preference.

For more information, please refer to Suspension by Harris Warburton

Aerodynamics

The aerodynamic package on the car is justified by a potential 117.7 points gain over a car without one. It consists of a front wing, rear wing, undertray and diffuser, nosecone and bodywork. Based on the performance of HEV2 (2019 GDP), the goal was to match its downforce of 360 N, while improving the downforce-to-drag ratio (L/D). A target drag of 150 N was therefore set.

The front and rear wing designs aimed to maximise downforce and L/D . A double-element front wing configuration was selected over three-elements across its span to minimise drag. Rectangular endplates with rounded edges, as well as gurney flaps, were added to optimise the aerodynamic performance. The rear wing comprises a three-element design flanked by two tapered endplates spanning the inner width of the inboard part of the rear tyres and is placed at the rearmost and highest position allowed by the Formula Student (FS) regulations. For simplicity, both wings will undertake the same manufacturing process and will be made of carbon fibre reinforced polymer (CFRP). Aluminium rods from each wing will be directly attached to the chassis and are designed to maximise structural integrity and satisfy deflection constraints specified in the FS regulations.

The undertray and diffuser was the most efficient aerodynamic device with an L/D of 8.75. It incorporates an underbody rake of 2.2° , which was made possible due to the chassis upsweep. The diffuser design features strakes on the outer section to manage the tyre wake vortices and seal the main underbody flow, maximising the downforce generated. Using a similar manufacturing process as the wings, the undertray-diffuser will be made of CFRP. It will be mounted to the chassis using pipe clamps to limit deflection to within the bounds set by the FS regulations. The nosecone and front bodywork were jointly designed with the 3D CAD Modelling Team and will be made of glass fibre reinforced polymer (GFRP) to minimise cost. Computational Fluid Dynamics (CFD) analysis was performed to minimise flow separation and ensure clean flow to the sidepods. The CFD set up was validated against experimental data for the Ahmed Body, with the CFD result within 5.3% of accepted experimental results for C_D .

A full car CFD simulation was performed to understand the aerodynamic interaction between components as well as determine the aerodynamic balance of the car. The total downforce and drag were (301 ± 14) N and (145 ± 2) N respectively. The centre of pressure (CoP), normalised by wheelbase, was 16% aft of the centre of gravity (CG). With the present lapsim model, a negligible points loss was predicted compared to when CoP was forward of its present position. Although this car did not meet the set downforce target, a 90.9 point gain over a car without an aerodynamic package was still achieved, with the drag target being met. The aerodynamic package has a total mass of 27.1 kg, costing £6429 and is within the budget set by the team.

For more information, please refer to Undertray and Diffuser Design by Isaac Balbolia, Front Wing Design by Adrian-Andrei Buda, Rear Wing Design by Anna Renshaw, and Nose Cone, CFD Validation and Full Car Sim by Benjamin See

Ergonomics

The ergonomics was considered as past designs neglected the ergonomic aspects. This involved optimising the driver's environment to ensure overall peak performance for the vehicle would be attained. Joint angles were researched and experimented to find the optimal positioning of the driver to ensure the effects of fatigue were mitigated as much as possible. Upright and reclined back seat angles were investigated due to the advantages and disadvantages to both designs. The positioning of subsystems affecting the driver was explored to ensure human factors was acknowledged.

For more information, please refer to Ergonomics by William Ho

Cost and Manufacturing Analysis

Costing was pursued in a manner where initial sub teams would be allocated a budget. Cheapest options were taken for components that were deemed to have no impact on the performance of the car, with the mass also taken into consideration. Any components that would drastically increase the cost would first be examined to see whether the additional points gain to price ratio was within an acceptable range. The final cost of the vehicle was found to be £30,863.00.

For more information, please refer to Cost and Manufacturing Analysis by Jo Lee Lum

Steering and Braking

Any vehicle intending to complete a course with corners requires a steering system. For HEV4 the rules were studied to design a steering system that would produce optimum performance in the competition. Due to a requirement by regulation of a mechanical system to steer the front wheels rather than an electrical one, it is not possible to select a particular Ackermann percentage. Rather, the steering system was designed in a way that would an optimum average value over the events during the competition, in this case 82%. Many components were selected from commercial producers, however some were chosen to be custom-designed in order to ensure that they minimised stress and deflection whilst remaining no heavier than absolutely necessary, such as the tie rods and steering column straight sections. This

also allowed for setup adjustability to be possible with HEV4, making it more adaptable to different courses and track conditions.

The braking system was designed by initially determining a desired braking deceleration for the car. A deceleration of 1.5g from braking was determined to be a safe braking deceleration for a driver in this racing class. The braking system was designed to be hydraulically actuated, outlined in the Formula student rules to be a requirement, where disc brakes were chosen. A braking force of 1767.4N and 706.3N were determined to be the point at which the tyres would lock up, accounting for rolling friction. All components were selected to be bought from commercial producers as they provided HEV4 with the required braking performance, and stayed within the weight limits while maintaining some level of adjustability.

For more information, please refer to Steering System Design by Alexander Dorricott and Braking Systems Design and Integration by Nicholas Yap

3D Computer Aided Design

This year the 3D CAD sub-team played two important roles in the project: it supported the project by modelling the majority of the components of the car, and then it relayed key issues encountered when assembling the various components of the car to the respective sub-teams, collaborating with them to solve these problems. As expected, there were many obstacles to overcome when assembling the car however these kept to a minimum due to the proactive approach taken by the sub-team, such as being immediately aware of any parts approved to be modelled and discerning any geometric constraints that may arise due to the other sub-teams' designs, and even creating a conceptual car assembly early on in the project to better understand how the packaging of the car would look like. The 3D CAD sub-team was heavily involved in the packaging and layout of the car, and designed many mounts for the components that required them in order for them to be fixed to the chassis and not just float in space. There was also a lot of care and consideration put into the detail of the car and its components so as to make the most accurate representation of what the final car could look like if it were ever manufactured.

For more information, please refer to 3D Modelling and Packaging, Cockpit, Firewall, Nose and Bodywork Design. Final Assembly and Integration by Joshua Rodgers, Modelling of the Driver, Drivetrain, Wheels, and Braking System by Adrian Saletnik, and 3D Modelling by Matthew Tanggai Freddy

Powertrain

A holistic approach was taken this year when selecting the hybrid configuration, with a final choice of the through-the-road setup. This was found to minimise the possible packaging and centre of gravity issues that would result in a larger and uncontrollable car. Two through-the-road strategies were investigated: motor & ICE running together and separately. The separate strategy consisted of a motor ratio of 16 and drive ratio of 6 whereas the together strategy implemented a torque coupling method which utilised a motor ratio of 4. LapSim results showed a total score difference of 14.13 in dynamic events making the separate strategy the final design.

A motorcycle engine, Triumph Daytona 675R with a theoretical peak power of 93.6 kW, and mass of 70 kg was chosen due to its highest power-mass ratio, and reasonable cost amongst all the others. In-hub motor configuration was chosen by having two APS12090 brushless direct current (BLDC) motors with a peak power of 18kW and stall torque of 100.46 Nm due to its high instant power and torque at low speeds for starting. With compatible voltage and current requirements to the motor, the two motor controllers chosen are MST 140-200. To fulfil the power requirements of the motors and the energy requirements of the endurance event, a total of 96 Murata VTC6 cells are chosen, placed in a 16-cell in series and 6-cell in parallel arrangement.

Advice was taken on from previous years to investigate the possibility of including a supercharged or turbocharged engine. The turbocharger was found to have no overall points effect due to it only

improving performance at high speeds which would not be reached during the race. A supercharger was found to increase performance but would require significant redesign of the engine which was not possible in the time frame and budget allocated. Hence, other improvements such as the implementation of a custom exhaust system, ECU remapping and RPM cut-off controller were implemented. These improvements totalled to an additional 25.22 points. Overall, this resulted in an improvement in predicted position of first place.

For more information, please refer to Powertrain Development by Adem Bououdine, Battery Design and Powertrain Optimisation by Shuting Wang, and Powertrain Selection and Together Strategy by Zimeng Amber Jia

Electrical System and Powertrain Integration

The role of Electrical System and Powertrain Integration consists of two parts. Electrical System focuses on the selection of critical electrical components. The target of electrical system is to ensure safety and monitor vehicle performance during the race. Therefore, a data logger AiM MyChron 5S, an Orion 2 168 Cell Battery Management System (BMS), a water temperature sensor, an oil pressure sensor, and a throttle position sensor are selected. The cost, weight and power consumption are evaluated and shortlisted. Position of sensors, wiring and fusing are considered and discussed with 3D CAD Modelling Team and Ergonomic, Packaging, Layout, Cost Team to facilitate the layout and ensure no interference will occur. Powertrain Integration consists of the design of a new battery pack to house all high voltage battery cells. An improved battery pack design based on 2021 EV21 battery pack is implemented with BMS kit. A passive water-cooling system is included after discussion with Powertrain Team and Cooling Team. A brief drawing of BMS voltage tap locations is included to illustrate the safety measures if any error is detected. Further improvements of the electrical system include the calibration of the sensors and hardware programming, which are left incomplete due to GDP time constraints. Further simulations and tests should be done to examine the battery pack performance under different weather conditions.

For more information, please refer to Electrical System & Powertrain Integration by Qinyi Mia Wan

Cooling

The endurance event within the competition sees the car run at its limit for over 30 minutes and this makes the temperature rise of key components a critical concern, especially due to the hybrid nature of the vehicle. The engine, electric motors, motor controllers, battery and brake discs were identified as high risk components due to the impact of overheating on vehicle performance. The electric motors and battery employed a passive liquid cooling approach in which they were packed with a volume of water that sufficiently increased their heat capacity to prevent overheating despite ignoring heat loss to the surroundings. Two air-cooled motor controllers were implemented but due to competition regulations they could not be placed in a location with adequate ventilation. Instead each motor controller was placed on an aluminium plate in the floor enabling sufficient convective cooling due to the high underfloor velocities. The grey cast iron brake discs were adequately cooled by a combination of convective and radiative cooling, remaining below the stated operating maximum of 550°C. The engine was the highest power component on the car and was designed by the manufacturer to be liquid-cooled. A cooling circuit was implemented containing an expansion tank and reservoir, an electric pump and two side-mounted radiators in a parallel configuration and was able to keep the engine operating within its optimal temperature range. To improve the mass flow through the radiators, basic composite sidepods were designed based on experimental data and implemented as one with the floor and bodywork.

For more information, please refer to Cooling System Design by Ishaan Roy

1 Introduction

The purpose of this Group Design Project is to develop the prototype of HEV4, a fully-compliant combustion-electric hybrid car design for the Imperial Racing Green team, capable of competing in the 2022 Formula Student competition and aiming to maximise the achievable points. The need of an aerodynamic package on a Formula Student car was widely questioned in recent years, with numerous teams managing to win the competition despite not using any aerodynamic devices. However, data from past GDP reports and previous winning car designs, as well as results from the lap simulator, have proved the suitability of an aerodynamic package able to provide a significant gain in points for a minimal penalty in weight and cost. Its main functionality is to generate downforce, improving the traction and handling capabilities of the car and increasing the cornering speeds for faster lap times, hence being particularly useful in the endurance event. This report outlines the design process of the front wing, an essential aerodynamic device providing up to 35-40% of the total downforce of the car [1], with a key role in ensuring the aerodynamic balance, diverting the flow over the tyres, as well as providing clean air to the floor and diffuser and to the cooling system, hence increasing their performance.

2 Compliance with regulations

The most important aspect within the design stage of the front wing is to ensure that all the regulations imposed for the 2022 Formula Student competition, found in [3], are fully satisfied. These are related to both the maximum sizes allowed for the aerodynamic devices, as well as their structural integrity. All the rules affecting directly the front wing design are summarised below:

- The front wing should not extend more than 700 mm in front of the foremost point on the tyres and should allow a 75 mm clearance in front of the above-mentioned point to avoid any impact with the steering tyre. The height relative to the ground is limited to 250 mm for the sections positioned directly in front of the tyres and 500 mm for any other section of the front wing.
- All edges that are prone to impacting with a pedestrian must have a minimum radius of 5 mm for the horizontal ones and 3 mm for the vertical ones, respectively.
- The front wing should not deflect more than 10 mm in the main load-carrying direction when a 200 N force is applied over a minimum surface of 225 cm² and more than 25 mm in any direction, at any point, when it is loaded by a 50 N force.

3 Baseline Configuration

3.1 Targets and Layout Selection

The main objective in the development of the front wing was to maximise the performance not only of the wing itself, but of the wing in relation to other components of the car. On the one hand, the design process of the front wing aimed to reach the aerodynamic targets, namely 141 N of downforce and 37 N of drag, explained in more detail in the GDP 2022 "Undertray and Diffuser Design" report [2]. Moreover, a sensitivity study was carried out by the Vehicle Dynamics team, as highlighted in the GDP 2022 "Vehicle Performance Simulation" report. This was used to quantify how each alteration in downforce, drag or mass affects the points achieved, based on the slopes in Table 1, where dP , dL , dD and dM are the change in points, downforce, drag and mass, respectively [4].

Table 1: Sensitivity study slopes.

dP/dL [points/N]	dP/dD [points/N]	dP/dM [points/kg]
0.4628	-0.2732	-0.3900

On the other hand, all design decisions studied the potential impact on other components of the car, therefore multiple layouts have been analysed within the baseline configuration stage. Comparing designs adopted by past reports and Formula Student winners [5] [6] [7], it was decided to split the front wing into three different sections along the width and maximise the distance between the tyre external edges, without exceeding the track limits.

The inner section was restricted by the presence of the nose cone, hence could not accommodate more than one wing element, neither horizontally nor vertically. Therefore, the only driving criterion of its design was to ensure it could be safely placed under the nose, leaving a moderate gap for the flow to be efficiently driven to the floor and diffuser assembly and not severely deflected, hence improving their performance. The outer section, located in front of the tyres, was limited to a maximum height of 250 mm and could not fit more than two wing elements without reducing the aerofoil chords and compromising the performance. Moreover, its key role in diverting the air efficiently over the spinning tyres made a single-element configuration unfeasible, hence a double-element layout was selected for the outer section. The middle section was not as restricted and could accommodate both double and triple-elements configurations, as long as they complied with the length restriction of 625 mm while allowing significant clearance for endplates. This section directly affected the amount of mass flow available for the cooling system and therefore had to maintain the wake as low as possible. The layouts can be visualised in Figures 8a and 8b of Section 4.3.1. When double-element configurations were selected for both the middle and outer sections, the aerofoil sections were connected and the setting angles preserved so that to remove the endplate and create a continuous wing element comprising the two.

3.2 Aerofoil Selection

As mentioned previously, the driving criteria when designing the front wing were to maximise the level of downforce generated, to minimise the drag produced through a high lift-to-drag ratio, as well as to ensure that any stalling behaviour is avoided. The performance of several aerofoils was studied using the open-source XFLR5 software and by setting the Reynolds number to 4.5×10^5 for the main element and 2×10^5 for the flap, respectively. These values have been selected based on an average cornering speed of 15 m/s provided by the lap simulator [4] and an average chord of 450 mm for the main aerofoil and 200 mm for the flap, in line with previous GDP reports [5] [6] [7]. The plots of lift coefficient and lift-to-drag ratio against angle of attack for the main aerofoil are displayed in Figures 1a and 1b, while the data for flaps are highlighted in Figures 12a and 12b in Appendix A.

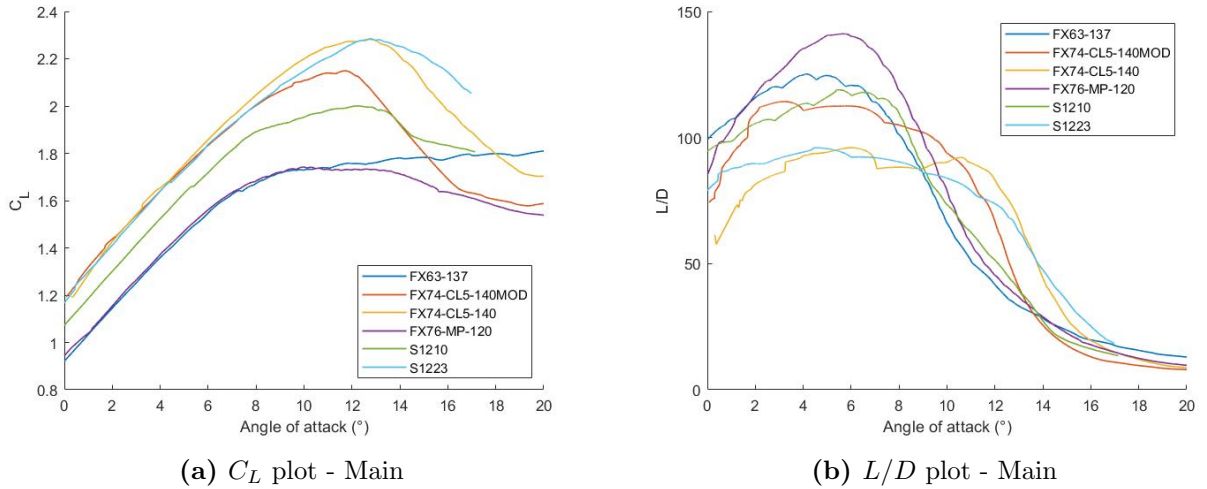


Figure 1: Aerofoil data for main element

For the main element, the six aerofoils have been shortlisted based on their high-lift characteristics at low speeds, as well as a thickness-to-chord ratio between 10% and 14% to ensure a trade-off between

structural integrity, good stall characteristics and low levels of drag generated. Whilst the angle of attack required for maximising the lift-to-drag ratio could be selected between 4° and 6° , a higher incidence provided more downforce and was desirable. Therefore, the main aerofoil angle of attack was set to 6° and the best lifting performance at this incidence was observed for S1223 and standard and modified FX74-CL5-140, respectively. However, all these three were seen to have very thin sections close to the trailing edge and could not be properly manufactured using carbon fibre sheets for such small-sized wings. The compromise solution was the FX76-MP-120 aerofoil that provided the highest lift-to-drag ratio of all sections analysed, but with a significant lift coefficient penalty. One of its advantages was the more flat pressure surface that could reduce the flow deflection behind the wing and drive the flow efficiently to the floor and diffuser, increasing their performance. Furthermore, the maximum thickness position was found between 30% and 34% of the chord, providing ideal stall characteristics for low-Reynolds number applications [8]. A sectional profile of the FX76-MP-120 aerofoil is shown in Figure 2a.

With regards to the flaps, it was decided to use the same aerofoil for all elements to facilitate the manufacturing process and save costs. All candidates have been chosen for their excellent lifting capabilities at low speeds and delayed stall at angles of attack higher than 20° . The aerofoil most suitable for the front wing configuration was considered the smoothed CH-10. This yielded the second best performance in terms of lift coefficient and lift-to-drag ratio at moderate incidences between 15° and 30° corresponding to the first flap, as well as an acceptable level of downforce at very high angles of attack, over 40° , for the second flap. A sectional view of the smoothed CH-10 profile is highlighted in Figure 2b below. The two aerofoils have also been tested at a minimum speed of 10 m/s and a maximum one of 40 m/s, showing almost no change in the lift coefficient and a maximum variance of $\pm 8\%$ in the lift-to-drag ratio.

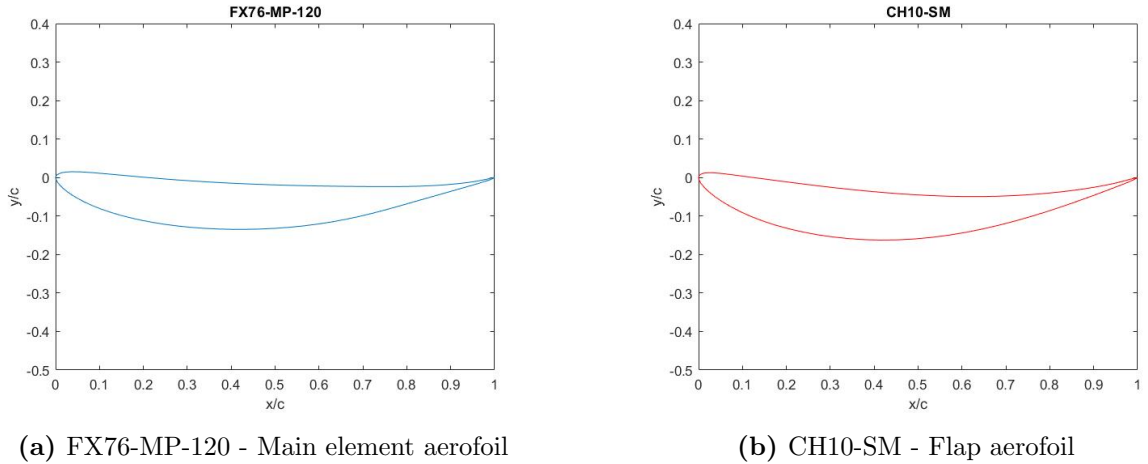


Figure 2: Aerofoil profiles selected for the main wing and flaps

4 Design and Optimisation using CFD

In general, the design process of aerodynamic devices in the car racing industry is performed using a combination of analytical, computational and experimental methods, in order to optimise each parameter for a range of environmental conditions and to predict as accurately as possible the resulting flowfield and the expected performance. Due to the limited timeframe of the project, performing wind tunnel testing was not a viable option and it was decided to use CFD simulations as the main tool in the optimisation process, with the aid of the Star-CCM+ software.

It was preferred to determine the ideal values of parameters such as setting angles, chords, gaps, overlaps and ground clearance using two-dimensional simulations of the flow, as these are not significantly affected by vorticity effects. Due to the time costly nature of three-dimensional simulations, these were

only used for testing overall configurations of the front wing once the main parameters have been set, as well as to study the effects of endplates and additional flow devices aiming to reduce the effects of vorticity and separation and increase the performance.

4.1 2D CFD Simulations

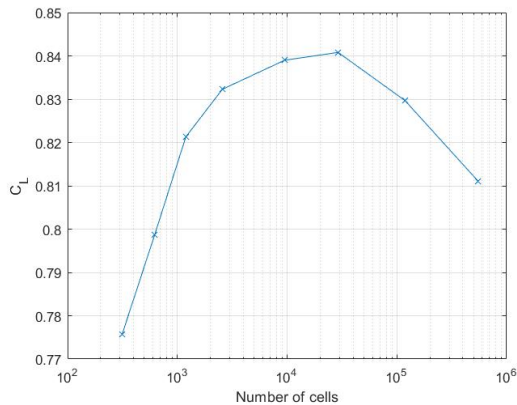
4.1.1 Setup

The setup selected for 2D simulations of each section aimed to ensure a trade-off between accuracy and computational time. A mesh convergence study was carried out, as it will be presented in the next section, in order to find the maximum base size to yield reliable results and minimise the time required for the significant number of simulations. Further refinements of specific areas, such as the leading and trailing edges of the aerofoils, were necessary for better capturing the high pressure gradients in these regions. Due to the time constraints of this preliminary phase of the design, it was preferred to use a steady, segregated, constant density flow model for the front wing, considering it is exposed to a clean and steady flow, without any negative effects from other components. The setup was preserved for the simulations with integrated tyres, despite the unsteady nature of the flowfield produced by their rotation, as it was assumed that the 2D simulations are not massively affected.

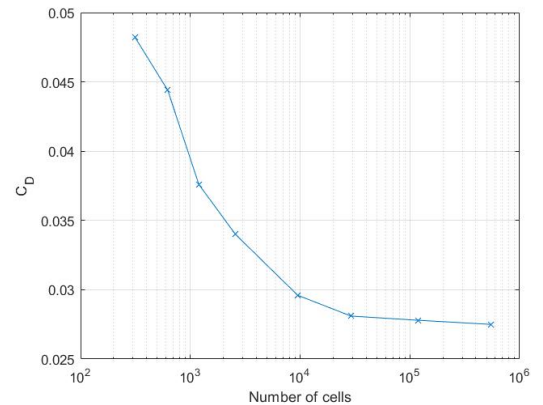
The domain was set to ensure that the flow at the outlet reaches the freestream conditions, therefore requiring 5 chord lengths upstream the wing and 10 chord lengths downstream. The height was set to 8 chord lengths, with a 44 mm ground clearance below the lowest point of the main aerofoil. The inlet velocity and outlet pressure were chosen as 15 m/s and 1 atm, respectively. For accurately capturing the behaviour of the wing section in real conditions, the ground was modelled as a moving floor, with its relative velocity also equal to 15 m/s. Two-dimensional simulations have been run for each section along the width of the wing, in all configurations presented previously. Therefore, three setups have been created: two for the middle section, one with three aerofoil elements and one with two aerofoil elements, as well as one for the outer section in front of the tyre, where only a double-aerofoil configuration complied with the height restriction. For the latter, a rotating tyre model was adopted, with a rotating speed set at 57.5 rad/s.

4.1.2 Mesh Convergence

The FX76-MP-120 aerofoil was also used to perform a mesh convergence study of the setup, by decreasing the base size for the simulations from 0.5 m and 314 cells to as low as 0.002 m and 548458 cells. The plots in Figures 3a and 3b below depict the change in lift and drag coefficients with increasing number of cells and refining the mesh.



(a) Lift coefficient vs. Number of cells plot



(b) Drag coefficient vs. Number of cells plot

Figure 3: Mesh convergence analysis of 2D simulation

Although the C_L values increased for a number of cells up to 30000, a decreasing trend was seen above that value due to a more accurate prediction of turbulence and an associated drop in lift. However, the C_D plot clearly showed a converging trend for a number of cells higher than 120000 and a base size lower than 0.005 m, therefore this was the value decided to be used in all two-dimensional simulations for ensuring reliable results. An example of the two-dimensional mesh applied to one of the configurations is highlighted in Figure 13a in Appendix A.

4.1.3 Mesh Validation

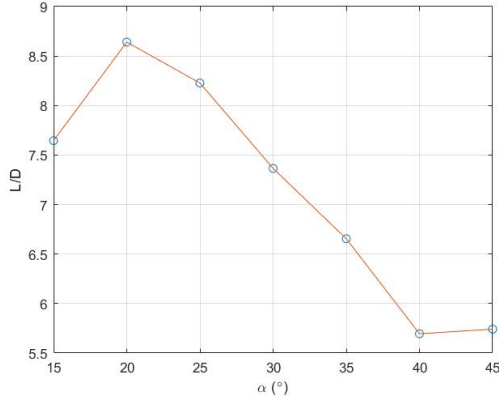
In order to check the validity of the 2D simulations performed, a benchmark test was designed using the main aerofoil selected, set at different angles of attack for the same Reynolds number as in the XFLR5 estimations ($Re=4\times10^5$). The $K-\omega$ SST turbulence model was compared to the $K-\epsilon$ one, the former being more accurate in capturing the near-wall behaviour, while the latter is widely used for simulating only freestream flows. As the $K-\omega$ SST model combines characteristics of both $K-\omega$ and $K-\epsilon$, it was expected to be the most suitable in the analysis of the front wing. For curved surfaces such as aerofoils, it was desired to predict as accurately as possible the turbulence in the viscous sub-layer, therefore the non-dimensional wall distance y^+ had to be kept lower than 1 for the $K-\omega$ SST model to predict correctly real flow conditions. The benchmark test was performed at incidences between 0° and 12° for both $K-\omega$ SST and $K-\epsilon$ and the resulting lift coefficient value for each angle of attack was compared to the data obtained in XFLR5, as highlighted in Figure 14 in Appendix A. Below 9° , both turbulence models have underestimated the amount of lift generated, however the $K-\omega$ SST predicted higher C_L values, much closer to the ones from XFLR5. Moreover, considering that the solver behind the XFLR5 software could not accurately capture any turbulence and vorticity occurring over the aerofoil, the two turbulent models in Star-CCM+ delayed the stall angle of attack of the aerofoil. Therefore, based on its numerous advantages and better predictions, the $K-\omega$ SST turbulence model was selected for all 2D and 3D simulations.

4.2 Parameter Optimisation

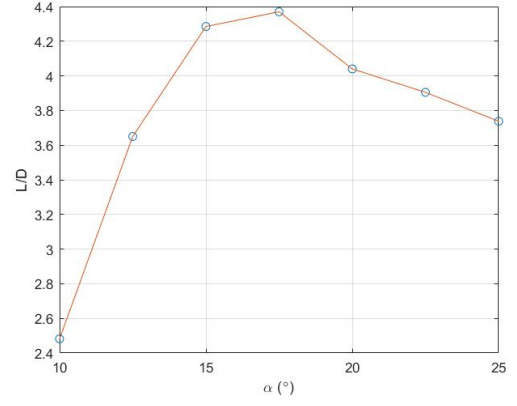
The main objective of the 2D simulations was to get the best set of wing parameters, such as chord ratios or flap incidences, that could maximise the lift-to-drag ratio for each configuration. In order to obtain an optimal design, it was decided to study the effect of changing each parameter on the sectional wing performance, while keeping all the others constant, due to the limited time available. Therefore, the setting angles of the flaps in both two-elements and three-elements configurations have been selected by iterating through the range of possible values that satisfy the regulations, with all other parameters set at values that were thought to help achieving the maximum performance available. A ground clearance of 44 mm, approximately 10% of the main chord, as well as a constant slot gap of 13.5 mm (3% of the main chord) and an overlap distance of 18 mm (4% of the main chord) were selected [9]. An ideal main chord to flap chord ratio c_m/c_f was set to 70:30, testing a range of total chords to maximise the space available and comply with the allowable sizes [10]. Therefore, the horizontal total chord was picked equal to 610 mm, in order to allow for the addition of endplates and fit into the 625 mm length restriction on the front wing. A MATLAB code has been created to visualise each configuration and check its validity within the height and length limitations of each region.

4.2.1 Double-Element Configuration

Firstly, the main wing element was tested with a single flap, ensuring the chord ratio is satisfied by setting the chords to 440 mm and 190 mm, respectively. Two simulation setups have been considered: one with a tyre behind the wing, for which the more restrictive height limitation allowed for a flap angle between 10° and 25° to fall under the rules, as well as one without the tyre, that did not limit the range of incidences. The main element setting angle was kept at 6° for maximising the performance. The lift-to-drag ratio L/D was recorded for each test and the values plotted in Figures 4a and 4b with increasing flap setting angle.



(a) Configuration without tyre



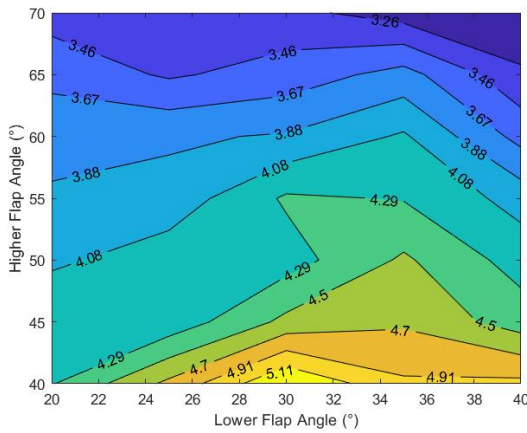
(b) Configuration with tyre

Figure 4: L/D against flap setting angle α

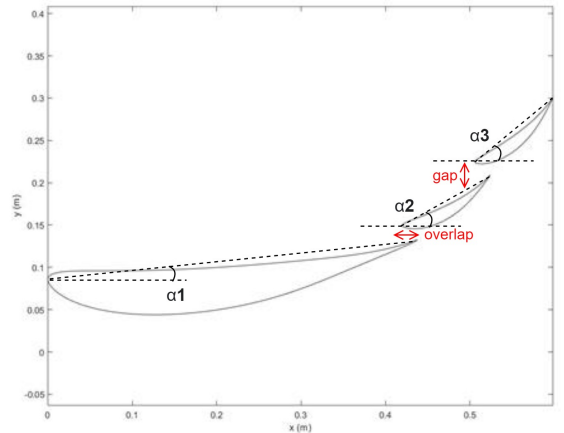
As observed in the plots, the highest L/D for the simulations that included the tyre was recorded at a setting angle of 17.5° , while the ones without a tyre yielded a maximum L/D at 20° incidence. As the wider section without a tyre behind was found to contribute more to the aerodynamic performance of the front wing, the setting angle for the two-elements configuration was fixed at 20° , resulting into a lift-to-drag value of 8.64.

4.2.2 Triple-Element Configuration

The second configuration to be analysed consisted of a main element and two flaps of equal chord lengths, with the aim of maintaining the chord ratio between the major element and the flaps assembly equal to 70:30. Therefore, a 440:120:120 mm chords configuration was found to maximise the space within the regulation box while keeping the main element as long as possible, as a main contributor to generating downforce. Moreover, the slot gap and overlap values between consecutive elements have been kept constant and in line with the two-element configuration. Simulations have been carried out to sweep through angles between 20° and 40° for the lower flap and between 40° and 70° for the higher one. Any higher incidences highlighted an undesirable amount of separation over the aerofoil sections and a significant drop in performance. The lift-to-drag values have been recorded for each test and a contour plot of all results has been created, as observed in Figure 5a below. A visualisation of the general triple-element setup is displayed in Figure 5b.



(a) Contour plot of L/D



(b) Layout visualisation

Figure 5: Triple-element configuration

As depicted in the plot above, the lift-to-drag ratio was maximised for a lower flap angle of 30° and a higher flap angle of 40° . The flow over this layout was seen to stay attached over most of the three

aerofoils surfaces, while ensuring clean air to the cooling system. Therefore, the best triple-element configuration yielded a maximum lift-to-drag ratio of 5.32.

4.2.3 Chords ratio

As mentioned previously, a chords ratio of the main and flap element equal to 70:30 has shown the best performance in past experiments on low-Reynolds high-lift generating wings [10]. However, it was decided to test several layouts, with a percentage of the main element chord relative to total chord ranging between 60% and 80%, as indicated by the plot in Figure 6. The simulations results confirmed the choice of a 70:30 ratio as being the most suitable for the front wing design, as this yielded the highest lift-to-drag ratio.

4.2.4 Slot Gap and Overlap

The effect of varying the size of the slot gap and overlap was studied only within the double-element configuration, sweeping over values between 2% and 4% of the main chord for both parameters, at the incidences found to maximise the performance. The results are highlighted in Figure 7 below. A gap of 3%, or 13.5 mm, was considered an excellent trade-off between a wider wake generating high amounts of drag and a very close proximity between the aerofoil elements that may induce boundary layer separation, as seen in Figure 7. The overlap was maximised in order to reduce the drag by ensuring a smoother flow on the suction section. Therefore, a 4% value, or 18 mm, was considered ideal, as an amount of overlapping higher than that was observed to cause separation on the flap's leading edge.

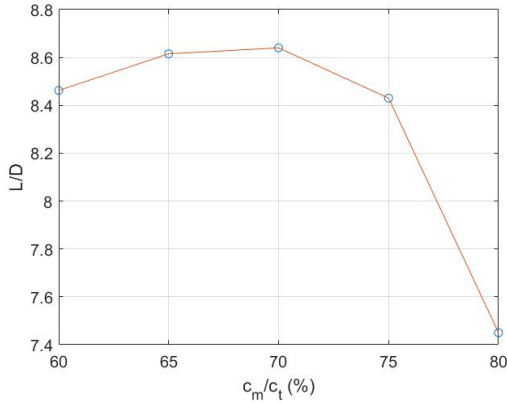


Figure 6: L/D for varying chords ratio.

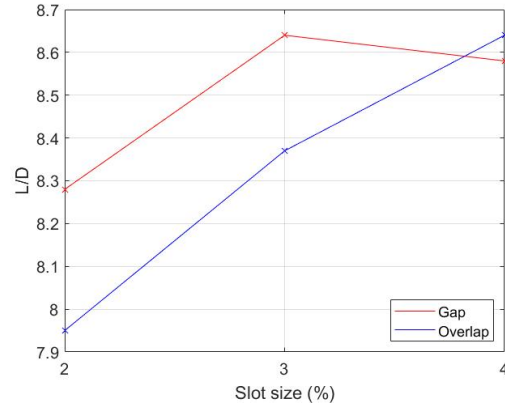


Figure 7: L/D for varying slot sizes.

4.2.5 Overview of 2D simulation results

Overall, two-dimensional simulations significantly helped in achieving the best layouts that could maximise the performance of the front wing within the restrictions imposed, through monitoring the effects of changing each parameter involved in the design in the shortest time possible. The parameters yielded by the most efficient double-element and triple-element configurations are summarised in Table 2, clearly highlighting that the double-element configuration ensured a lower L/D value and was more suitable. However, to confirm this result, testing the two wing layouts using 3D simulations was considered necessary.

Table 2: Summary of 2D simulations

Configuration	Main Element [°]	First Flap [°]	Second Flap [°]	C_L	C_D	L/D
2-Elements	6	20	-	2.87	0.33	8.64
3-Elements	6	30	40	3.20	0.64	5.32

4.3 3D CFD Simulations

Once the list of parameters has been finalised for all sections studied, it was decided to perform three-dimensional simulations for major decisions surrounding the design. Each step of the decision-making process was labelled as an "iteration", with the points gain resulting from each update on the design being quantified based on the sensitivity analysis previously presented in Section 3.1. Taking into account the computationally-expensive nature of 3D simulations and the limited timeframe of the project, only a few configurations have been tested. The analysis focused on studying the negative effects of turbulence and vorticity on the real wing's performance, as well as the ability of endplates and other flow devices to diminish these effects. Only half of the wing has been used, so that to halve the computational costs, with the vertical mid-plane being set as a symmetry plane in the setup. The domain has been set in a similar manner as for the 2D simulations, with a moving floor implemented. Despite the significant effect of rotating tyres on the wing performance, it was decided not to include them in the analysis due to the requirement of an unsteady solver to accurately predict the instabilities induced, which was unfeasible given the computational power available. Therefore, the impact of tyres was studied only within the full car simulations. The global base size was selected equal to 0.3 m, as a trade-off between saving computational cost and improving the results' accuracy. Surface controls have been preferred for the aerofoil sections, along with a localised finer mesh of 0.002 m base size, as well as 15 prism layers required to precisely predict the flow behaviour in the viscous sub-layers along the aerofoils curvature. A comparison between the 2D and the 3D mesh applied is shown in Figures 13a and 13b in Appendix A.

4.3.1 First iteration - Number of elements

The width of the front wing has been selected to maximise the available front track width, without exceeding the track limits, therefore to be equal to 1582 mm without extending outboard of the tyre edge. The inner section of the wing, placed under the nose, had a 450 mm width imposed by the bodywork design, with the endplate thickness equal to 8 mm to ensure the structure is strong enough. The outer section was 542 mm wide on each side, divided into 232 mm for the two-elements section and 310 mm for the three-elements section in the triple-aerofoil configuration.

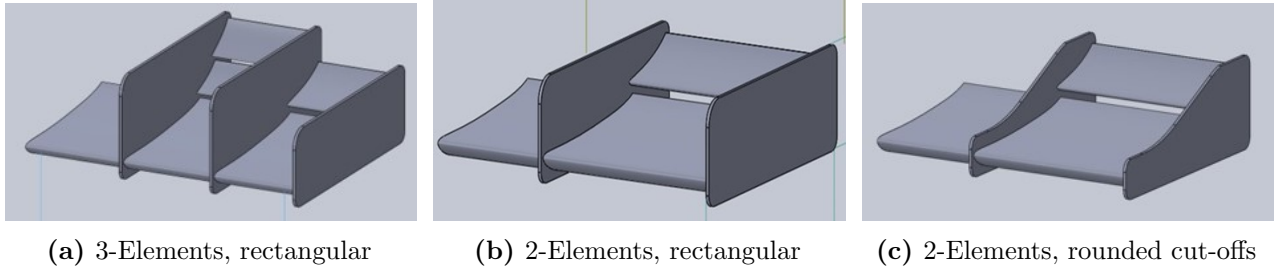


Figure 8: Comparison between configurations tested.

The first iteration aimed to compare the double-element with the triple-element configuration, using a standard, rectangular endplate design as highlighted in Figures 8a and 8b, so that to confirm the results from 2D simulations. The outputs of the analysis are included in Table 3, clearly showing the significant advantage provided by the 2-elements configuration: it generated a higher downforce for a lower drag, along with reducing the weight added to the design. Moreover, it was found to reduce the height of the wake, ensuring clean air and a sufficient mass flow rate to the sidepods, as well as to contribute more efficiently to the braking system cooling. Therefore, the double-element configuration was selected for the whole width of the wing, except under the nose cone.

Table 3: Two-elements compared to three-elements layout.

Configuration	Downforce [N]	Drag [N]	L/D	Change in points
2-Elements	174.60	28.98	6.02	+6.5
3-Elements	167.65	39.54	4.24	-

4.3.2 Second iteration - Endplates

The second iteration studied how endplates could be refined in order to maximise their contribution to the aerodynamic performance of the front wing. Two designs have been developed: a 625x210 mm standard, rectangular endplate with rounded corners, as well as a "trapezoidal one" with rounded cut-offs, as shown in Figures 8b and 8c. All endplates were provided with rounded edges of at least 3 mm and 5 mm radius, respectively, to comply with the regulations. The objective was to maximise the allowed length and height so that to efficiently seal the edges of the aerofoil sections and avoid the formation of tip vortices, leaving enough margin above and below the surface of each aerofoil. The results of the analysis are summarised in Table 4, showing that the implementation of cut-offs reduced the mass of the design, but came with a significant penalty in lift-to-drag ratio that did not justify the weight saved. Hence, the rectangular endplates were retained on the front wing based on the significant points gain predicted.

Table 4: Comparison of endplates designs: rectangular and rounded cut-offs.

Endplate design	Downforce [N]	Drag [N]	L/D	Change in points
Rectangular	174.60	28.98	6.02	+3.2
Rounded cut-offs	165.80	31.56	5.25	-

4.3.3 Third iteration - Additional flow devices

After the second iteration, a new nose cone design required the front wing middle section to be widened from 450 to 550 mm. A simulation was run for the new layout, with the results highlighted in Table 5, showing a 8.7% decrease in downforce, a 10.5% reduction in lift-to-drag ratio and an associated drop of 6.7 points. For the third iteration of the design process, two widely used flow devices have been proposed, which have demonstrated their positive impact on the aerodynamic performance in the racing industry: Gurney flaps and footplates. Some other proposals have been analysed, such as vents, curved endplates or winglets, however their addition was considered unfeasible and too complex based on the limited time and resources available to accurately predict their positive contribution. The Gurney flap was attached only to the flap of the outer section, as it can be observed in Figure 15 in Appendix A, because it did not fit below the nose cone without negatively affecting the flowfield. The size was decided based on past experimental data, showing that a height between 0.5% and 2% of the flap chord provides enough performance improvement [11]. Therefore, the height was set at 3.8 mm, or 2% of the chord, to provide enough deviation of the flow over the tyres, with a thickness of 1.5 mm to ensure it is manufacturable. The footplate was attached to the exterior endplate of the front wing, as shown in Figure 16 in Appendix A. The diameter was set to 32 mm, in order to allow enough gap for the flow to pass without compromising the space for the aerofoil sections, considering that the width was limited to 1582 mm and the front wing could not extend outboard of the tyres. The two flow devices have been tested in separate simulations, as they were thought not to affect each other's performance. The results of the two simulations are presented in Table 5 below, along with the alteration in points provided by each design when compared to the standard model.

Table 5: Comparison of Gurney flap and footplate configurations with the standard wing.

Configuration	Downforce [N]	Drag [N]	L/D	Change in points
Standard	159.40	29.6	5.39	-
Gurney flap	165.50	32.05	5.16	+2.2
Footplate	157.50	29.30	5.37	-0.9

As indicated in the table, the Gurney flap provided a significant improvement in the design, with a 2.2 points increase. Although the lift-to-drag performance slightly decreased, it was desired to keep the Gurney flap in order to increase the downforce generated and get closer to the aerodynamic targets set. Moreover, the Gurney flap was found to deviate the flow over the tyres more efficiently, while

decreasing the separation on the suction side of the aerofoil. As regards the footplate, it could not prove its real benefit at the low speeds the car is operating at and only added complexity to the design. Therefore, as it took away useful space from the aerofoil sections and slightly decreased the downforce generated by the wing for a similar amount of drag, it was decided not to use footplates in future iterations.

4.3.4 Fourth iteration - Pitch test and Final Configuration

Another alteration in the nose cone design due to a change in the tyre position required a further increase in inner section width from 550 mm to 650 mm, along with a reduction in the main element chord from 440 mm to 360 mm for the same section. The new performance of the wing is summarised in Table 6, highlighting a significant drop in downforce and drag compared to the previous iteration, with a slight improvement in lift-to-drag ratio. Gurney flaps have been then added to this design, with the downforce, drag and lift-to-drag ratio values for the final configuration of the front wing highlighted in yellow in Table 6 below.

Table 6: Performance data of final front wing configuration (with and without Gurney flaps)

Configuration	Downforce (N)	Drag (N)	L/D	Change in points
Standard	145.40	25.48	5.71	-
Gurney flaps	158.40	28.90	5.47	+5.1

The last step of the analysis was to check the robustness of the design, by testing the front wing in maximum pitch down condition and ensure no stalling behaviour is observed. For maximum braking and average aerodynamic loads acting on the front wing, a maximum pitch down angle of 1° was found, resulting in a 35 mm maximum vertical travel for the front wing. A safety margin of 5 mm was applied to ensure the wing does not touch the ground under extreme loading conditions or unpredictable torsion behaviour, therefore the endplates were placed at 40 mm above the ground, with the aerofoil sections moved from 44 to 47.5 mm in order to maintain a similar performance for the endplates. The pitch test highlighted a 7.5% drop in downforce to 146.4 N, considered reasonable, hence the front wing configuration was validated.

5 Full Car Simulations

CFD simulations have also been performed on the full car model that included the final configuration of the front wing. Their aim was to study the overall performance of the car and compare it to the results of each component tested in isolation, as well as to study all the interactions that exist between components from an aerodynamics viewpoint, as explained in more detail in GDP 2022 "Nose Cone, CFD Validation and Full Car Simulation" report [12]. The performance of the front wing has diminished considerably relative to individual testing, as expected, due to the presence of the nose cone and the spinning tyres in its proximity. The rotation of the tyres opposing the direction of the flow created a low speed, turbulent region over the front wing flap, notably decreasing both downforce and drag generated. A comparison between data from individual testing and the full car simulation is presented in Table 7, highlighting a similar drop in both downforce and drag, but maintaining an almost constant lift-to-drag ratio, a good marker of the consistency of front wing's efficiency.

Table 7: Comparison between performance in individual testing and full car simulations.

Configuration	Downforce [N]	Drag [N]	L/D	$SC_L [m^2]$	$SC_D [m^2]$	$\Delta SC_L [\%]$	$\Delta SC_D [\%]$
Individual	158.40	28.90	5.47	1.19	0.22	-	-
Full car	119.00	23.00	5.17	0.89	0.17	-25.21	-22.72

Moreover, a visualisation of the streamlines showing the flow behaviour over the front wing is displayed in Figure 10 of Section 8, with an additional sectional velocity scene displayed in Figure 17 of Appendix

A. The data from full car simulations were considered a more reliable and accurate indicator of the front wing's performance under track conditions.

6 Material Selection and Structural Integrity

For selecting the most suitable material for manufacturing the front wing, the CES Edupack software was used to generate an adequate Ashby chart, displayed in Figure 18 of Appendix A. The wing was idealised as a bending panel, therefore a suitable merit index to be plotted against cost, a key parameter in the design, was $E^{1/3}/\rho$ [13], where E is the Young's modulus and ρ is the material density, in order to ensure a light and stiff structure. As materials such as metals were considered unfeasible, the two best candidates were carbon fibre and glass fibre reinforced polymer (CFRP and GFRP), the former being much stronger and more expensive than the latter. Several past designs have opted for CFRP with epoxy resin as the best solution for the front wing, mainly based on its ability to significantly reduce the weight of the front wing while ensuring a very smooth surface for the aerofoil sections to avoid any separation induced.

The XC130 2x2 Twill 3K Prepreg Carbon fibre was selected based on its low mass per square metre (210 gsm) and will require 8 layers of 0.25 mm each [14], arranged in a $[\pm 45^\circ, 0^\circ, 90^\circ]$ layup to be able to sustain the bending stresses. The $\pm 45^\circ$ outer layers will be added to protect the core of the composite structure from debris, given the proximity of the front wing to the ground. The inside of the wing element will be made of a polystyrene block, to minimise the weight, cut in the aerofoil shape required. Wax will be needed to stick the first carbon fibre sheet to the polystyrene core, then the epoxy resin will be applied between every two consecutive sheets, placed in the configuration previously-mentioned. The manufacturing process will follow closely the steps presented in the GDP 2022 Rear Wing Design report [15].

Based on the choice of materials, a cost and mass estimation was carried out in collaboration with the Ergonomics, Layout and Cost team, resulting in a total mass as low as 4.55 kg, with an associated production cost of 1470£ [16]. An FE analysis was conducted to check the validity of this selection, considering the two loading cases depicted in the regulations. The results can be visualised in Figure 9 below, showing a maximum deflection equal to 0.45 mm for a pressure load applied over the whole wing surface and to 1 mm for a point load acting on the endplate, significantly lower than the maximum allowable deflection in the most restricting scenario.

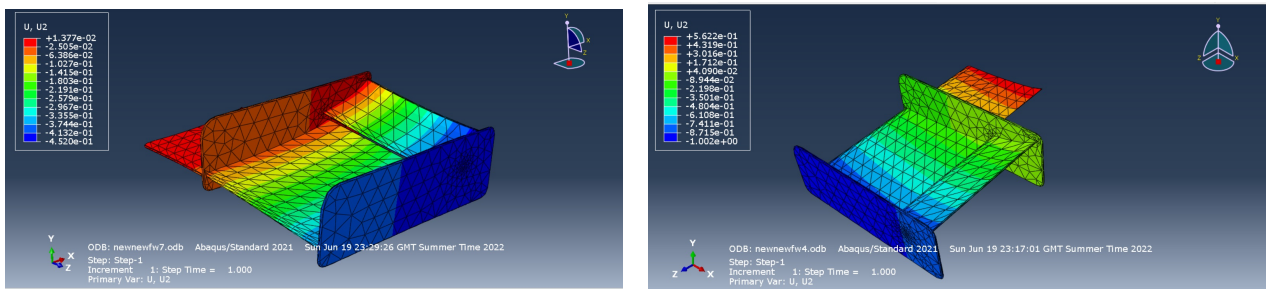


Figure 9: Deflection plot of the front wing under pressure (left) and concentrated loading (right).

7 Mounting System

A mounting system for the front wing was also designed, with the aim to connect the aerodynamic device to the rest of the car by attaching it to the hard points in the front area of the chassis, without severely affecting the flowfield around the nose cone. It was decided to use two sets of cylindrical mounting rods made of Aluminium 8090, in line with similar elements attached to the chassis. One

pair was designed to counteract the aerodynamic loads acting at the centre of pressure of the main wing element, connecting it to the higher hard points on the chassis. Each rod will integrate an aluminium plate to be bolted into the wing element, as well as small, rectangular, lateral extrusions on the chassis to ensure the mounting rod doesn't interfere with the impact attenuator. The second pair was designed to restrict any sideways motion or pitch rotation of the front wing, as well as to avoid excessive deflections on the thinner flap element, connecting the flaps and endplates to the lower hard points on the chassis. The mounting mechanism is shown in Figure 11 of Section 8.

The main pair of rods was sized to resist any bending, buckling and shear failure. The most restrictive was the buckling condition, that yielded a minimum allowed diameter of 5.6 mm for the cylindrical rods, using a safety factor of 2 applied to the maximum aerodynamic and inertial loads encountered. However, a 10 mm diameter was selected to ensure that no failure occurs due to any unexpected torsional behaviour on the wing. It was also decided to keep the same size for the second set of rods, so that to facilitate the manufacturing process.

8 Evaluation and Conclusion

Overall, the front wing configuration designed performed well in reducing the drag compared to previous proposals, however with a significant penalty in the downforce generated, expected in a double-element configuration with low incidences. The 119 N of downforce achieved only 84% of the targeted performance, while the drag of 23 N induced by the front wing addition was 38% lower than estimated, overall managing to maintain an efficient lift-to-drag ratio of 5.17. The added mass of the front wing was only 4.55 kg, mainly as a result of using CFRP, for an induced cost of 1470£. An overview of the main parameters, as well as of the aerodynamic performance, can be found in Tables 8 and 9 in Appendix B. An overall visualisation of the front wing is found in Figure 11 below, while a technical drawing is displayed in Figure 19 of Appendix C. As a starting point for future iterations of the design, more complex features may be analysed in order to assess their impact on performance, such as twisted wing elements, curved endplates, vents or winglets attached to the endplates. Moreover, wind tunnel tests may prove useful in validating the computational results achieved through CFD simulations. To conclude, the addition of an aerodynamic package was justifiable and led to a 90.9 points increase when compared to a design without aerodynamic devices, with a 39.0 points contribution solely from the front wing [2].

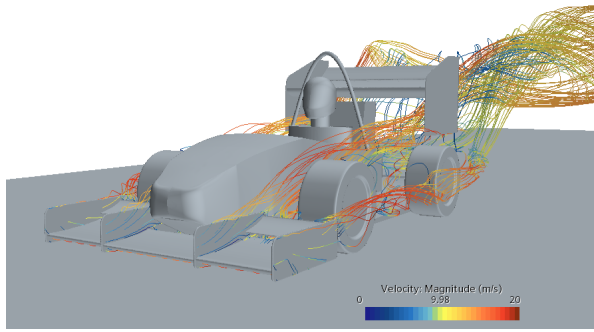


Figure 10: Streamlines visualisation [12].

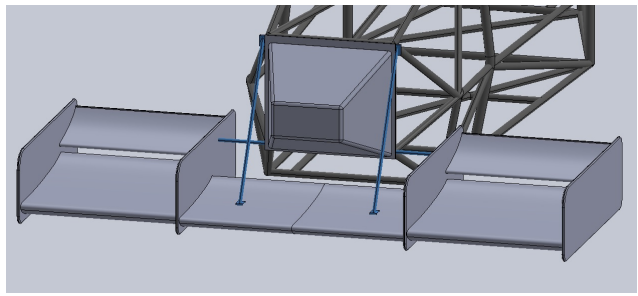


Figure 11: Front wing and mounting system.

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9 Appendix A: Figures

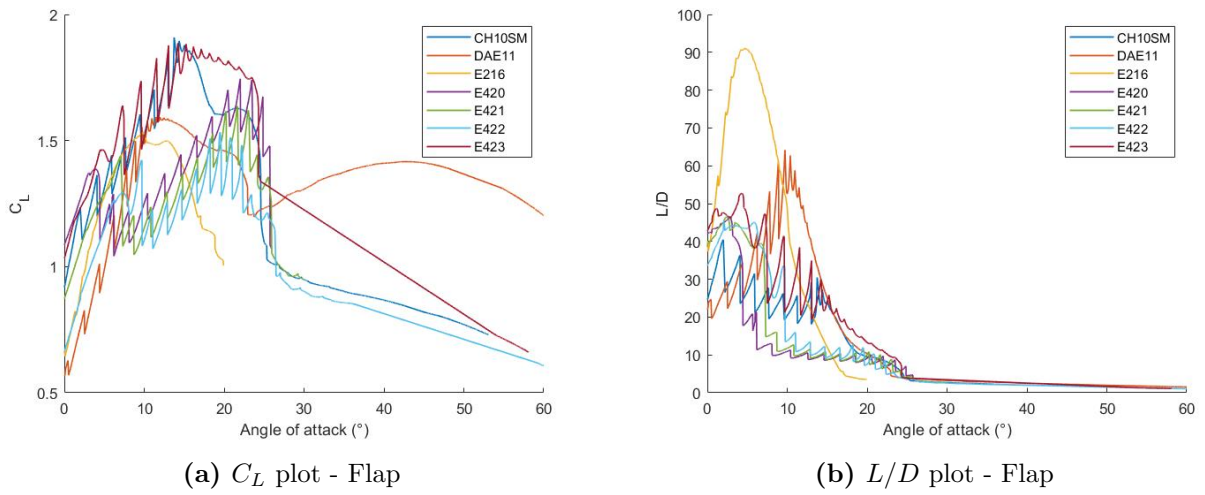


Figure 12: Aerofoil data for the flap element

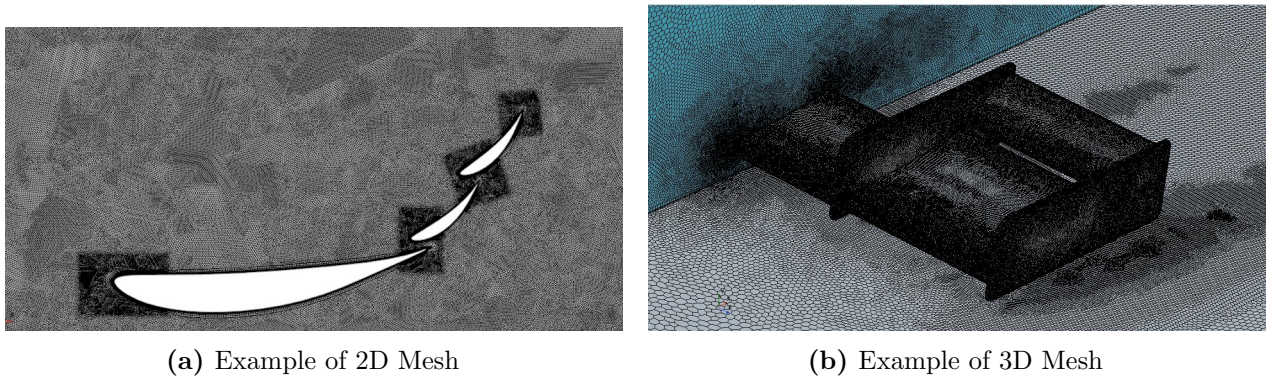


Figure 13: Comparison between two-dimensional and three-dimensional mesh.

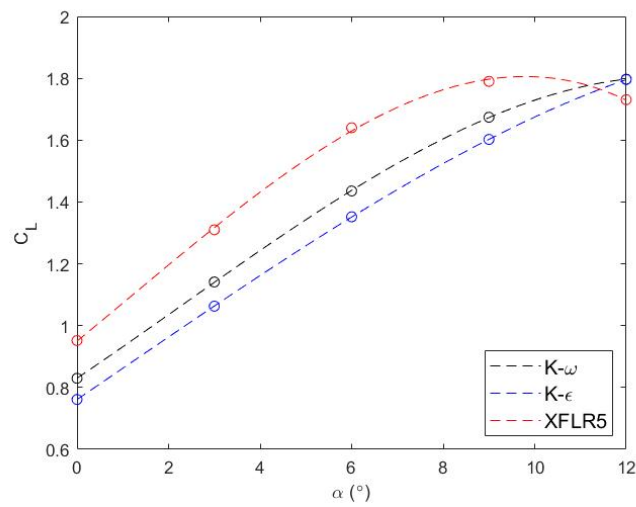


Figure 14: Comparison of turbulence models and XFLR5 data

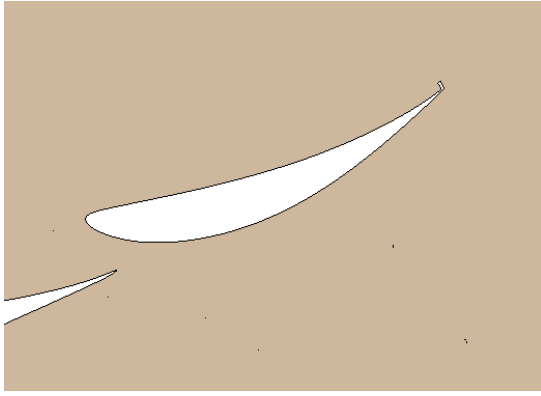


Figure 15: Flap and Gurney

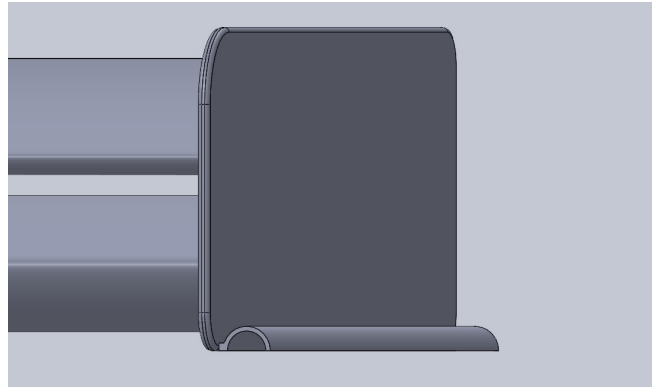


Figure 16: Footplate connected to the endplate.

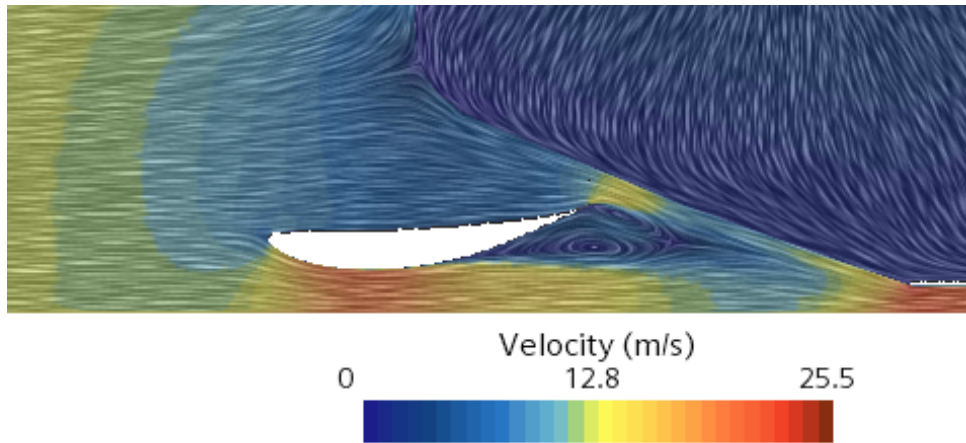


Figure 17: Sectional vector scene of the full car simulation, focusing on the interaction between nose cone and front wing inner element [12].

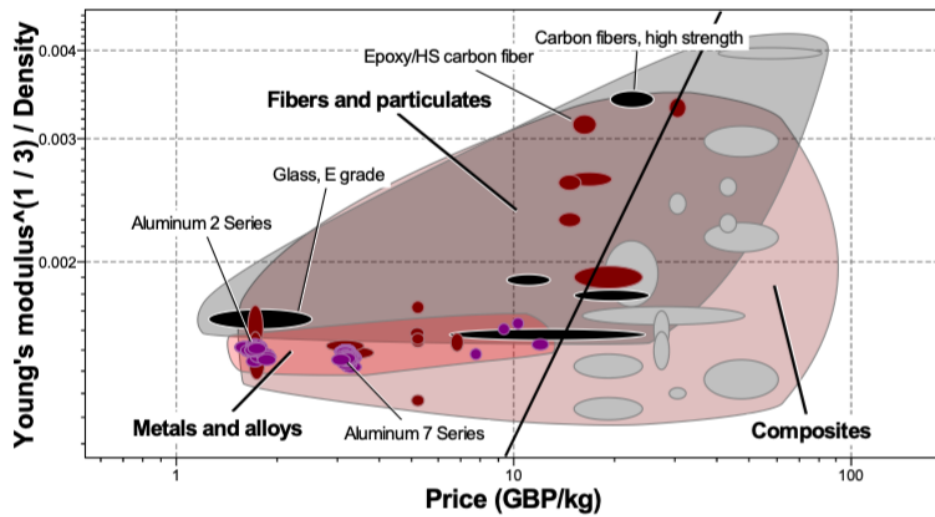


Figure 18: Ashby chart used for the material selection process [15].

10 Appendix B: Tables

Table 8: Summary of Front wing parameters

Parameter	Value
Configuration	Double-element
Main aerofoil	FX76-MP-120
Main setting angle	6°
Main chord (inner section)	360 mm
Main chord (outer section)	440 mm
Flap aerofoil	CH-10SM
Flap setting angle	20°
Flap chord	190 mm
Total width	1582 mm
Inner section width	650 mm
Outer section width	900 mm
Endplate thickness	8 mm
Endplate height	210 mm
Endplate length	625 mm
Endplate vertical edge radius	3 mm
Endplate horizontal edge radius	25 mm & 10 mm
Gurney flap thickness	1.5 mm
Gurney flap height	3.8 mm
Diagonal mounting rod length	495.39 mm
Horizontal mounting rod length	143 mm
Mounting rod diameter	10 mm

Table 9: Summary of Front wing performance

Parameter	Value
Downforce	119 N
Drag	23 N
L/D	5.17
Mass	4.55 kg
Cost	1470£
Points gain (FW only)	39.0
Points gain (aerodynamic package)	90.9

11 Appendix C: Technical Drawing

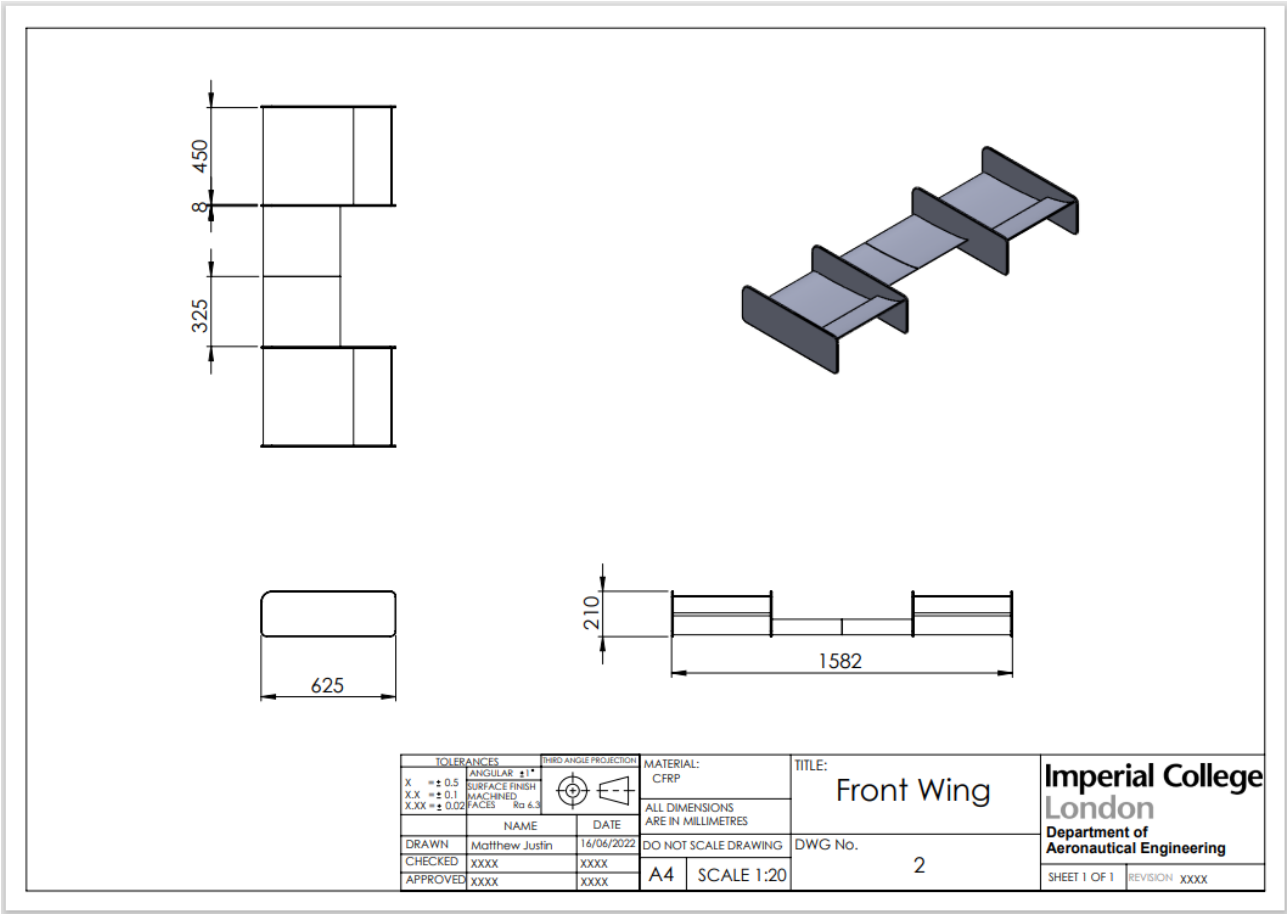


Figure 19: Technical Drawing of the Front Wing [17].